

# Follow the Gradient

# The power of differentiation

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- The big idea: optimisation by following gradients
- Recap: what are gradients and how do we find them?
- Jacobians as local linear maps
- Jacobian-vector and vector-Jacobian products

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  - How do we select those parameters?
- In deep learning/differentiable programming we typically define an objective function that we *minimise* (or *maximise*) with respect to those parameters
- This implies that we're looking for points at which the gradient of the objective function is zero w.r.t the parameters

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  - First order methods, second order methods, subgradient methods...
- With deep learning we're primarily interested in first-order methods<sup>1</sup>.
  - Primarily using variants of gradient descent: a function  $F(\mathbf{x})$  has a minima<sup>2</sup> (or a saddle-point) at a point  $\mathbf{x} = \mathbf{a}$  where  $\mathbf{a}$  is given by applying  $\mathbf{a}_{n+1} = \mathbf{a}_n - \alpha \nabla F(\mathbf{a}_n)$  until convergence from some initial point  $\mathbf{a}_0$ .

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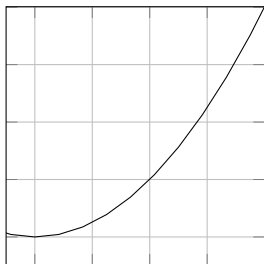
<sup>2</sup>not necessarily global or unique

# Recap: what are gradients and how do we find them?

## The derivative in 1D

- Recall that the gradient of a straight line is

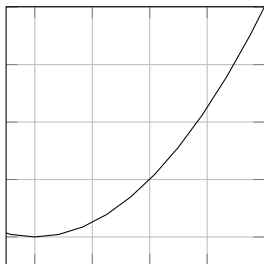
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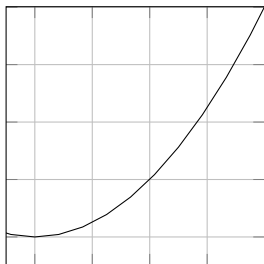
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- For an arbitrary real-valued function,  $f(a)$ , we can approximate the derivative,  $f'(a)$  using the gradient of the *secant line* defined by  $(a, f(a))$  and a point a small distance,  $h$ , away  $(a + h, f(a + h))$ :  $f'(a) \approx \frac{f(a+h) - f(a)}{h}$ .



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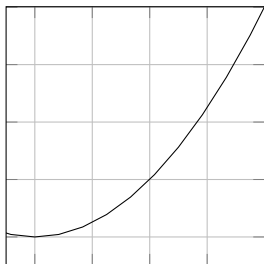
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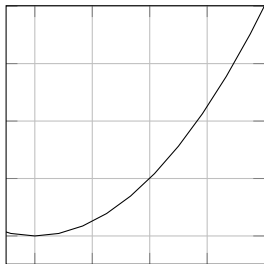


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- This expression is 'Newton's Quotient' or 'Fermat's Difference Quotient'.
- As  $h$  becomes smaller, the approximated derivative becomes more accurate.
- If we take the limit as  $h \rightarrow 0$ , then we have an exact expression for the derivative:  
$$\frac{df}{da} = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}.$$





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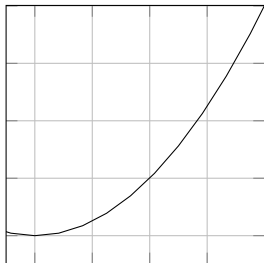
$$\frac{dy}{dx} = 2x$$

# Recap: what are gradients and how do we find them?

Aside: numerical approximation of the derivative

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- $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h}$

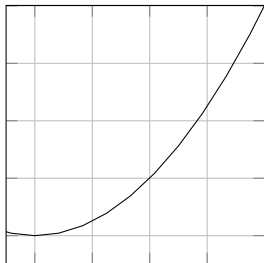


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- The bit inside the limit is known as the *symmetric difference quotient*



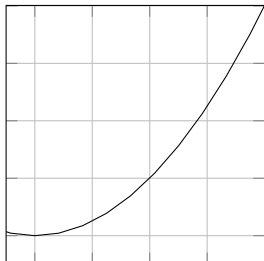


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- For small values of  $h$  this has less error than the standard one-sided difference quotient.



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Aside: numerical approximation of the derivative

- If you are going to use difference quotients to estimate derivatives you need to be aware of potential rounding errors due to floating point representations.
  - Calculating derivatives this way using less than 64-bit precision is rarely going to be useful. (Numbers are not represented exactly, so even if  $h$  is represented exactly,  $x + h$  will probably not be)
  - You need to pick an appropriate  $h$  - too small and the subtraction will have a large rounding error!

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## Derivatives of deeper functions

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  - e.g.  $z = f \circ g(x) = f(g(x))$

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- Deep learning is all about optimising deeper functions; functions that are compositions of other functions
  - e.g.  $z = f \circ g(x) = f(g(x))$
- The chain rule of calculus tells us how to differentiate compositions of functions:
  - $\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}$

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Example: differentiating  $z = x^4$

Note that this is a silly example that just serves to demonstrate the principle!

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Equivalently, from first principles:

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$$\frac{dz}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h}$$

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  - Equivalently,  $\mathbf{y}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{y}(t+h) - \mathbf{y}(t)}{h}$  if the limit exists.

# Recap: what are gradients and how do we find them?

Functions of multiple variables: partial differentiation

- What if the function we're trying to deal with has multiple variables<sup>3</sup> (e.g.  $f(x, y) = x^2 + xy + y^2$ )?
  - This expression has a pair of *partial derivatives*,  $\frac{\partial f}{\partial x} = 2x + y$  and  $\frac{\partial f}{\partial y} = x + 2y$ , computed by differentiating with respect to each variable  $x$  and  $y$  whilst holding the other(s) constant.

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  - This is the **gradient** of  $f$  at  $a$ .
- In the case of a vector-valued multivariate function, the partial derivatives form a matrix called the **Jacobian**.

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  - They involve multiple variables, which are often wrapped up in the form of vectors or matrices, and more generally tensors.
  - **How will we find the gradients of these?**

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## The chain rule for vectors

Suppose that  $\mathbf{x} \in \mathbb{R}^m$ ,  $\mathbf{y} \in \mathbb{R}^n$ ,  $g$  maps from  $\mathbb{R}^m$  to  $\mathbb{R}^n$  and  $f$  maps from  $\mathbb{R}^n$  to  $\mathbb{R}$ .

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If  $\mathbf{y} = g(\mathbf{x})$  and  $z = f(\mathbf{y})$ , then

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$$\frac{\partial z}{\partial x_i} = \sum_j \frac{\partial z}{\partial y_j} \frac{\partial y_j}{\partial x_i}.$$

Equivalently, in vector notation:

$$\nabla_{\mathbf{x}} z = \left( \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \right)^\top \nabla_{\mathbf{y}} z$$

where  $\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$  is the  $n \times m$  Jacobian matrix of  $g$ .



# Recap: what are gradients and how do we find them?

Example:  $\nabla_{\mathbf{W}} f(\mathbf{X}\mathbf{W})$

- Let  $\mathbf{D} = \mathbf{X}\mathbf{W}$  where the rows of  $\mathbf{X} \in \mathbb{R}^{n \times m}$  contain some fixed *features*, and  $\mathbf{W} \in \mathbb{R}^{m \times h}$  is a matrix of weights.
- Also let  $\mathcal{L} = f(\mathbf{D})$  be some scalar function of  $\mathbf{D}$  that we wish to minimise.

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- Also let  $\mathcal{L} = f(\mathbf{D})$  be some scalar function of  $\mathbf{D}$  that we wish to minimise.
- What are the derivatives of  $\mathcal{L}$  with respect to the weights  $\mathbf{W}$ ?

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Example:  $\nabla_w f(\mathbf{XW})$

- Start by considering a specific weight,  $W_{uv}$ :  $\frac{\partial \mathcal{L}}{\partial W_{uv}} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial W_{uv}}$ .

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- We know that  $\frac{\partial D_{ij}}{\partial W_{uv}} = 0$  if  $j \neq v$  because  $D_{ij}$  is the dot product of row  $i$  of  $\mathbf{X}$  and column  $j$  of  $\mathbf{W}$ .

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- We can then see that if we want this for all values of  $\mathbf{W}$  it simply generalises to:  $\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{X}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{D}}$ .

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  - ...but we don't want to repeat that index-heavy argument for every operation in a deep model!

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  - **The gradient of the loss with respect to a parameter tells you how much the loss will change with a small perturbation to that parameter.**

# Jacobians as local linear maps

Packaging up all of those partial derivatives

- For a function  $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ , the Jacobian collects all of the partial derivatives into an  $n \times m$  matrix:

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$$f(\mathbf{x} + \Delta\mathbf{x}) \approx f(\mathbf{x}) + J_f(\mathbf{x})\Delta\mathbf{x}.$$

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- So, the Jacobian tells us how a small input perturbation produces a small output perturbation.

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- These are *Jacobian-vector products* and *vector-Jacobian products*.



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  - We can compute this product without ever constructing the complete Jacobian.

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- A VJP takes sensitivities at the output and propagates them back to the input.



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- This is exactly the result we obtained from the element-wise derivation.

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The chain rule is composition of products

- If  $h = g \circ f$ , then a perturbation can be pushed forwards one operation at a time:

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  - Backpropagation (in the coming lectures) is a sequence of VJPs, starting at the loss and working backwards through the computation.



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- Later, we'll see how forward- and reverse-mode automatic differentiation compute these products through an arbitrary program.